

Lumova AI



Proposal for Building Production Platform

From AB Ark
to
Yancy

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1. Executive Summary

This proposal outlines the complete development roadmap to take Lumova.ai from its current proof-of-concept (POC) state to a fully live, investor-grade production platform hosted on AWS. The engagement is structured into three distinct phases, each independently fundable, so capital can be deployed in alignment with fundraising milestones.

The current POC has successfully demonstrated the core product concept: drone-captured data flows through a scoring engine to produce a transparent Building Health Score (0–100), a 3D Digital Twin with surface-anchored issue pins, a confidence publish gate, and a branded PDF dossier. The platform supports two user roles (Admin and Client) and is seeded with a Kansas City pilot portfolio of seven buildings.

What remains is the production engineering: a real backend on AWS, true multi-tenant data isolation, production authentication, live drone data ingestion, and the operational tooling that transforms this from a compelling demo into a billable SaaS product.

Phase	Scope	Investment
Phase 1 — MVP (Weeks 1–5)	Backend, auth, core API, AWS infra, CI/CD	\$8,000
Phase 2 — V1 AI Production (Weeks 6–14)	Real AI/CV, drone ingest, reporting, billing	\$20,000
Total Estimate	Full production platform	\$28,000

2. Current State — POC Assessment

The existing platform represents a high-quality frontend POC built with React/TypeScript. The following is an honest assessment of what is production-ready versus what requires re-engineering.

2.1 What is Working Well

- Building Health Score engine with auditable formula (Critical × 12, Moderate × 6, Minor × 2)
- AI Confidence Score with four weighted sub-metrics and 90% publish gate
- 3D Digital Twin with surface-anchored issue pins, orbit controls, and severity color-coding
- Two-role architecture: Admin panel (full CRUD) and Client portal (read-only)
- Branded PDF dossier generation with preliminary watermark logic
- Seven Kansas City pilot buildings across all major building types
- Mobile-responsive shell with off-canvas drawer and scroll-wrapped tables

2.2 What Must Be Re-Engineered for Production

- Data persistence: currently browser localStorage, no real database, no multi-tenancy
- Authentication: demo credentials only, no JWT, no session management, no SSO
- Drone data ingest: no pipeline exist, manual entry only
- AI/CV layer: simulated scoring, no real computer vision models
- Infrastructure: no AWS deployment, no CDN, no environment separation
- Security: no Row-Level Security, no tenant isolation at the data layer
- Billing and subscription: no payment infrastructure

3. Phase 1 — MVP Production Build

PHASE 1 MVP Production Build	Timeline Weeks 1 – 6	Investment \$8,000
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Phase 1 delivers a live, secure, multi-tenant production environment. At the end of this phase, Lumova.ai will run on AWS with real authentication, a real database, and a deployable backend API. The frontend POC will be migrated to production infrastructure.

3.1 Backend & API Layer

- Python FastAPI application with full REST and WebSocket support
- PostgreSQL on AWS RDS (db.t3.medium, Multi-AZ for production)
- Row-Level Security (RLS) policies for hard tenant isolation, clients can only access their own data
- JWT authentication with refresh token rotation and blacklist
- Role-based access control: Admin, Analyst, Client (read-only)
- Endpoints: buildings, analyses, issues, clients, users, reports, score engine

3.2 Authentication & Security

- AWS Cognito user pools with JWT validation at the API gateway
- MFA support (TOTP) for Admin and Analyst roles
- Separate staging and production environments with environment-specific secrets
- HTTPS enforced everywhere; HSTS headers; CSP policy

3.3 CI/CD Pipeline

- GitHub Actions: test → lint → build → deploy on merge to main
- Blue-green deployments via ECS Fargate, zero-downtime releases
- Automated database migrations with Alembic; rollback-safe
- Separate staging environment, all changes tested before production

3.4 Phase 1 Milestones & Payment

#	Deliverable	Amount
M1	AWS infrastructure provisioned, RDS live, S3 configured, environments separated	\$1,500
M2	FastAPI backend with all CRUD endpoints, RLS policies, Cognito auth integrated	\$2,200
M3	Frontend migrated to production AWS/CloudFront, login and role routing live	\$2,000
M4	Score engine API, PDF generation, CI/CD pipeline, staging sign-off	\$2,300
Total Phase 1 cost		\$8,000

4. Phase 2 — V1 Production Platform

PHASE 2 V1 Production Platform	Timeline Weeks 6 – 14	Investment \$20,000
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Phase 2 transforms the platform from a secured demo into a revenue-generating product. It introduces real AI/CV-based defect detection, an end-to-end drone data ingestion pipeline, advanced reporting, and a subscription billing layer. This is the phase that makes Lumova.ai fundable as a SaaS business.

4.1 AI / Computer Vision Layer

- YOLOv26 model fine-tuned on building defect imagery (cracks, spalling, corrosion, moisture staining, efflorescence)
- AWS SageMaker endpoint for GPU-accelerated inference, pay-per-call, scales to zero
- OpenCV pre-processing pipeline: image normalization, HDR merging, lens distortion correction
- Issue confidence scoring per detection, only detections above threshold are surfaced
- AWS Rekognition as fallback for general anomaly detection on initial ingestion
- Bounding box coordinates normalized to 3D twin surface coordinates via homography transform

4.2 Drone Data Ingestion Pipeline

- AWS Lambda-triggered ingest on S3 upload (supports DJI SRT telemetry, GeoTIFF orthomosaics, LiDAR .las files)
- Automated flight coverage calculator, measures what percentage of building surface was captured
- GPS coordinate extraction and mapping to building facade grid for pin placement

- Image quality scoring (resolution, focus, exposure) feeding the AI Confidence sub-score
- SQS queue for processing jobs, handles large ortho batches without timeout

4.3 Enhanced Reporting

- Server-side PDF generation with Puppeteer on Lambda, pixel-perfect Lumova branding
- Automated report scheduling: analysis complete triggers email with PDF link via SES
- Report versioning: each analysis creates a new immutable report version
- Client-accessible report history with download links

4.4 Billing & Subscription

- Stripe integration: subscription plans (per building, per analysis, enterprise flat-rate)
- Usage metering: analysis count tracked per tenant, enforced at API level
- Admin billing dashboard: MRR, churn, active subscriptions, overdue accounts
- Automated invoice generation and dunning logic via Stripe

4.5 Phase 2 Milestones & Payment

#	Deliverable	Amount
M5	AI/CV pipeline live on SageMaker, defect detection end-to-end tested	\$10,000
M6	Drone data ingestion pipeline (Lambda + S3 + SQS), flight coverage calculator	\$5,000
M7	Server-side PDF reports, SES email notifications, and report versioning	\$3,000
M8	Stripe billing integration, usage metering, and admin billing dashboard	\$2,000
Total Phase 2 cost		\$20,000

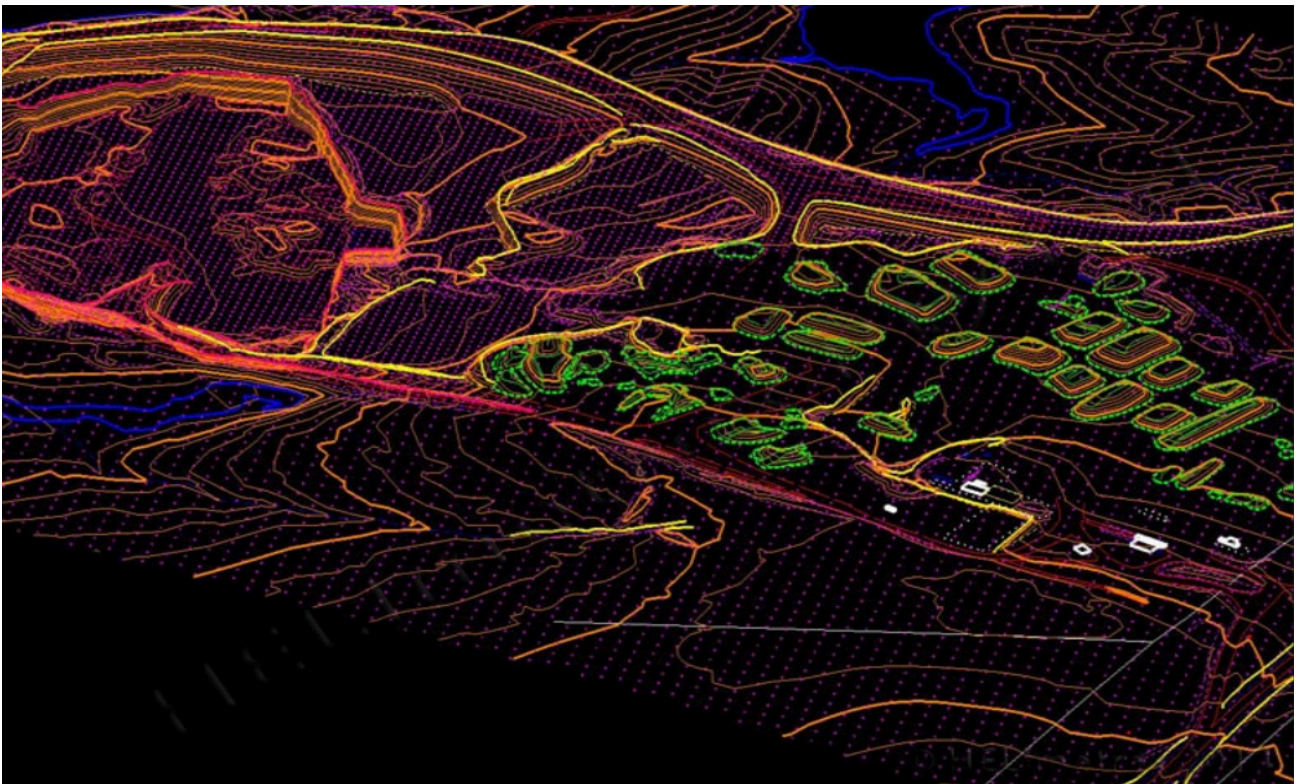
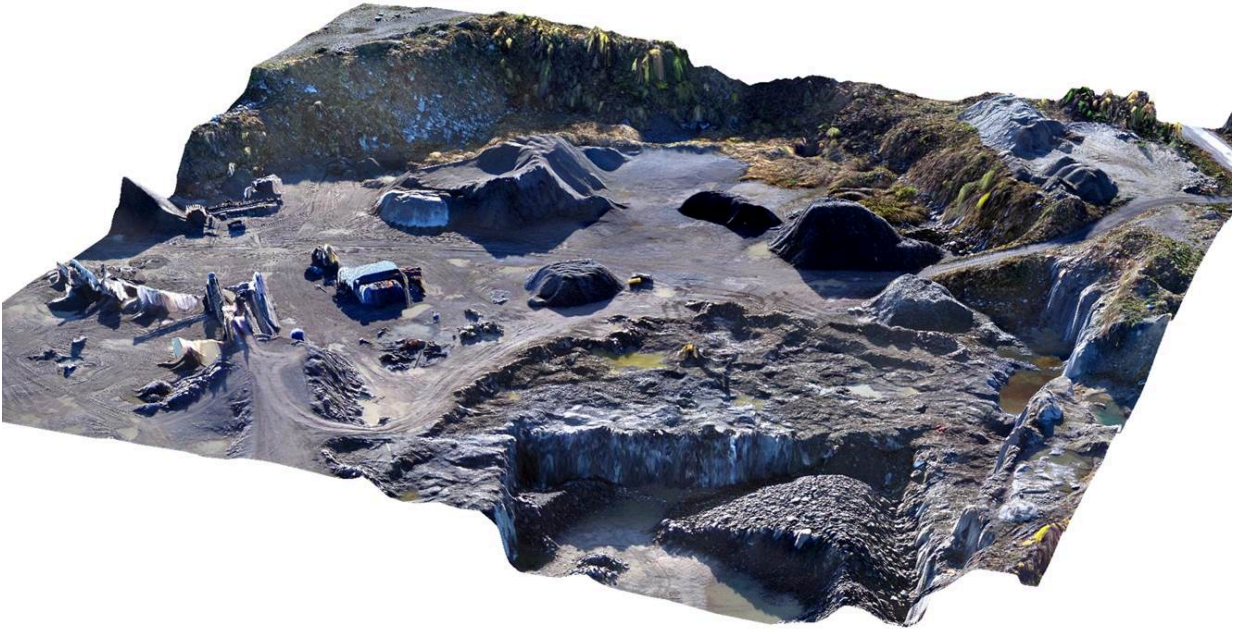
5. Future Envisioned Platform Features

PHASE 3 Advanced Platform	Timeline NA	Investment ~
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The Envisioned feature for Lumova is to establish The 3D Digital Frontier which is upgraded from a procedural model to photogrammetry-based mesh, historical trend analytics are introduced, a native mobile application is added, and enterprise authentication (SSO) enables Fortune 500 and municipal contracts.

5.1 Photogrammetry-Based Digital Frontier(3D - Image Meshing)

- Integration with drone-captured photogrammetry meshes
- Real building geometry replaces the current procedural stylized model
- Issue pins anchored to true GPS-referenced 3D coordinates from drone telemetry
- LOD (Level of Detail) streaming, mobile receives lower-poly mesh, Web receives full fidelity



5.2 Historical Trend Analytics

- Re-flight comparison: score delta between inspection dates, with issue-level change tracking

- Portfolio health dashboard: aggregate score trend across all buildings for a client
- Predictive maintenance suggestions based on issue progression rate
- Recharts-powered trend visualizations embedded in both portal and admin panel

5.3 Native Mobile Application

- React Native (Expo) **iOS and Android** apps for field technicians and client executives
- Offline-capable building viewer with cached twin geometry and issue list
- Push notifications for analysis completion and score threshold breaches
- Field photo upload directly to building S3 bucket from device camera

6. Full Technology Stack

The following stack is chosen for production-grade reliability, AWS-native integration, and alignment with the existing POC codebase. Deviation from the POC stack is minimized to protect the UI/UX investment already made.

Layer	Technology	Purpose
Frontend	React 19 + TypeScript + Vite 7	POC migration path, zero rewrite required
Routing	TanStack Router (file-based)	Type-safe routing, SSR-ready
State	TanStack Query + Zustand	Server state caching, UI state management
Styling	Tailwind CSS v4 + shadcn/ui	Existing design system preserved
3D	Three.js + @react-three/fiber	Digital twin, pins, orbit controls
Charts	Recharts	Trend analytics, score history
Backend	Python FastAPI + Uvicorn	Async REST API, WebSocket support
ORM	SQLAlchemy 2 + Alembic	Type-safe queries, migration management
Database	PostgreSQL 16 on AWS RDS	Multi-tenant with Row-Level Security
Auth	AWS Cognito + JWT	MFA, SSO, token rotation
Storage	AWS S3 + CloudFront	Imagery, PDFs, static assets
AI / CV	YOLOv26 on AWS SageMaker	Defect detection, GPU inference
CV Pre-proc	OpenCV + Pillow	Image normalization, homography
Ingest	AWS Lambda + SQS	Drone data pipeline, async job queue
Email	AWS SES + Jinja2	Transactional emails, report alerts
PDF	Puppeteer on Lambda	Server-side branded report generation

Layer	Technology	Purpose
Billing	Stripe + Stripe Webhooks	Subscriptions, usage metering, invoicing
Mobile	React Native (Expo)	iOS + Android (Phase 3)
IaC	Terraform + AWS CDK	Reproducible infrastructure, DR
CI/CD	GitHub Actions + ECS Fargate	Blue-green deploys, zero downtime
Observability	Grafana + Sentry + Datadog	Logs, errors, APM, alerting

7. Operational Cost

7.1 AWS Infrastructure Costs by Phase

Phase	Monthly AWS Cost	Notes
Phase 1 — MVP (1–10 buildings)	~\$410 / month	EC2 × 2, RDS, S3, CloudFront, ALB, SES
Phase 2 — V1 (10–50 buildings)	~\$850 / month	+ SageMaker inference, SQS, Lambda, Stripe webhooks
Phase 3 — Advanced (50+ buildings)	~\$1,600 / month	+ Enhanced SageMaker, RDS upgrade, CloudFront transfer

8. Engagement Model

8.1 Milestone-Based Payments

All phases are structured as milestone-based contracts. No phase is paid in full upfront. Each milestone has a defined, testable deliverable. Payment is released upon your acceptance of the deliverable, not upon a date.

- 25% of each phase paid at kickoff to cover infrastructure provisioning and setup costs
- Remaining 75% paid across deliverable milestones as defined in Sections 3–5
- Each milestone includes a 5-business-day acceptance window with a defined set of acceptance criteria
- Unused budget rolls to the next milestone if scope is delivered under estimate

8.2 Communication & Reporting

- Weekly async progress update (Loom video + written summary) every Friday
- Biweekly live check-in call (30 min) for demo, feedback, and priority alignment
- GitHub repository with branch-per-feature, full code visibility at all times

- Linear project board for task tracking, you will have read access throughout

8.3 What You Provide

- AWS account access (IAM user with appropriate permissions, or we provision under your account)
- Domain registrar access for lumova.ai DNS configuration
- Sample drone imagery and LiDAR files for the Kansas City pilot building
- Stripe account for billing integration (Phase 2)
- Timely feedback within the 5-business-day acceptance windows

8.4 Ongoing Retainer (Post-Delivery)

After Phase 3 delivery, an optional monthly retainer is available for:

- Infrastructure monitoring and AWS cost optimization
- Security patches and dependency updates
- Minor feature additions and client-requested tweaks
- On-call support for investor or client demonstrations

Retainer pricing: \$3,500 – \$6,000 / month depending on support tier.

9. Why This Partnership

9.1 Relevant Capabilities

Capability Area	Relevant Experience
SaaS Platforms	Multi-tenant SaaS products with subscription billing, role-based access control, and AWS-native infrastructure
AI & Computer Vision	YOLO-based object detection fine-tuning, SageMaker deployment, OpenCV pre-processing pipelines for industrial inspection
3D / WebGL	Three.js and @react-three/fiber applications including geospatial data visualization and interactive building models
Drone / GIS Data	Experience processing DJI telemetry, GeoTIFF orthomosaics, and LiDAR point clouds for inspection workflows
Investor-Ready Products	Delivered demo-ready platforms for pre-seed and seed-stage companies; understand the difference between a compelling demo and production software

9.2 Scoring System Improvement Suggestions

Based on review of the current formula, the following improvements would strengthen the scoring system for enterprise clients and insurers:

- Surface-area weighting: a crack spanning 40 linear feet should score worse than one spanning 4 inches, the current formula treats both as one Moderate issue
- Issue recency decay: issues detected in prior inspections that have not worsened should receive a small credit, rewarding maintenance engagement
- Structural vs. cosmetic classification: separate the score into a Structural Health sub-score and an Aesthetic Condition sub-score, insurers care only about the former
- Confidence-weighted score band: rather than a single number, surface a range (e.g., 74 ± 4) derived from the confidence sub-scores, more honest and more defensible in due diligence

10. Proposed Next Steps

#	Action	Owner
1	Review this proposal and share any questions or scope adjustments	Yancy
2	Schedule a 30-minute technical alignment call to walk through architecture decisions	Both
3	Agree on Phase 1 milestone definitions and acceptance criteria	Both
4	Execute Phase 1 agreement and 25% kickoff payment	Yancy
5	Provision AWS account access and share Kansas City pilot building assets	Yancy
6	Phase 1 development begins — Week 1 kickoff	AbArk

Lumova.ai is building something genuinely valuable, an auditable, visual, AI-powered building health platform that no one in the market delivers cleanly. The POC proves the concept compellingly. This proposal outlines the engineering work required to make that concept commercially real.

We are not proposing to rebuild what exists. We are proposing to put a production foundation under it, AWS infrastructure, real authentication, a real database, a real AI pipeline, so that every investor conversation and every client contract is backed by software that works the way it appears to work.

We look forward to building this with you.